Histopathology:

- Neoplastic **proliferation** of odontogenic epithelial cells in patterns:

Fancy People Always Get Benefits''

  - **F**ollicular type
  - **P**lexiform type
  - **A**canthomatous type
  - **G**ranular cell type
  - **B**asal cell type

Follicular Type:

- Most common.
- **Discrete follicles** of cells within fibrous connective tissue.
- Bordered by **tall columnar cells** (ameloblast-like).
- Central cells resemble **stellate reticulum**.
- **Microcyst formation** often seen.

Plexiform Type:

- Cells arranged in **irregular masses** or **interconnecting strands**.
- Masses bounded by **columnar cells**.
- Less prominent **stellate reticulum** than follicular type.
- **Cystic degeneration** in stroma common.

Acanthomatous type:

- Cells in the position of **stellate reticulum** undergo **squamous metaplasia** with **keratin formation**.
- **Epithelial or keratin pearls** may be observed.
- **Cystic degeneration** of connective tissue.
- **Ameloblast-like cells**.
- **Connective tissue stroma**.

Granular type:

- Transformation of **cytoplasm** to a **coarse granular eosinophilic** appearance.

Basal cell type:

- Resembles **basal cell carcinoma** of the skin.

Clinical Features:

- **Ameloblastoma** occurs between **10-90 years**.
- Average age of occurrence: **33-39 years**.
- More common in **males**.
- Affects the **mandible** more than the maxilla.
- Commonly found in the **molar-angle-ramus area** of the mandible.
- **Slow-growing, painless, bony hard swelling** of the jaw.
- May have **pain, paraesthesia,** and **mobility** of regional teeth.

Radiological Appearance:

- **Well-defined, multilocular radiolucent** area with a **honeycomb appearance**.
- Irregular and **scalloped margin**.

Differential Diagnosis:

- **Residual cyst**
- **Lateral periodontal cyst**
- **Giant cell granuloma**
- **Traumatic bone cyst**

Treatment:

- **Complete removal** of the tumor.
- **Curettage**.
- **Peripheral ostectomy**.
- **Intraoral block excision**.

- **Extraoral en bloc resection**.

- **Extraneous** – involving ascending ramus

Histological Features

- **Characteristic epithelial lining** with **parakeratin surface** (corrugated or wrinkled).
- Uniform thickness of epithelium (**6-10 cells deep**).
- **Palisaded, polarized basal cell layer** (picket fence or tombstone appearance).
- Sometimes shows **orthokeratin**.
- **Connective tissue** shows **satellite cysts** (high recurrence rate).
- Lumen may be filled with **straw-colored fluid** or **thick creamy material** (protein content **3.5 g%**).
- **No rete ridges** at the epithelium-connective tissue interface.

Differential Diagnosis

Syndrome Associated with OKC

- **Gorlin and Goltz Syndrome** (Jaw cyst, Basal cell nevus syndrome)
- Features:
    - **Multiple OKCs**
    - **Autosomal dominant** trait
    - **Basal cell carcinoma of skin**
    - **Frontal/temporal bossing**
    - **Hypertelorism**
    - **Mandibular prognathism**
    - Abnormal calcium/phosphate metabolism
    - **Calcification of falx cerebri.**

Indications for Biopsy

- When there is **doubt about a lesion** and it cannot be diagnosed clinically.
- When a lesion is in an **early stage** and shows only a superficial **ulceration** or **swelling**.
- When a lesion in the **oral cavity** does not heal.
- When a lesion **increases in size**, despite therapy.
- When the lesion is **unusual** and not diagnosable.
- Diagnosis by biopsy is particularly important in **early ulcerations** and **neoplasms**.

Types of Biopsy

- **Excisional Biopsy**: Entire lesion is excised for **histological evaluation**.
- **Incisional Biopsy**: Only a small section of tissue is removed from the lesion for **histological evaluation**.
- **Fine Needle Aspiration Cytology (FNAC)**: Tissue is aspirated from the lesion, and a smear is prepared for microscopic diagnosis.
- **Frozen Section Biopsy**: Performed to get an **immediate histological report** of a lesion. Tissue is frozen and sectioned for prompt diagnosis.
- **Punch Biopsy**: A circular piece of tissue is removed using a **punch** tool, especially useful for posterior **oropharyngeal region**.

Biopsy Technique

- **Paint the surface** of the area with **iodine** or a **coloured antiseptic**.

- Inject **0.5 to 2 ml of local anaesthetic** with adrenaline around the **periphery** of the lesion (not into the lesion).
- Use a **sharp scalpel** to avoid **tearing** tissue.
- If possible, **remove a border of normal tissue** with the specimen.
- Be careful not to **mutilate** the specimen when holding it with **forceps**.
- **Fix the tissue** immediately in **10% formalin** or **70% alcohol**. If the specimen is thin, place it on **glazed paper** before placing it in the fixative to prevent curling.
- Use a **scalpel** to remove a **4 to 8 mm piece** of the lesion and place it in a bottle with **10% formalin**.
- The biopsy specimen is sent to the **histopathologist** for diagnosis, ensuring it is **properly labelled**..

A benign **multifocal tumor of osteoclasts** seen in individuals over **40 years**. Characterized by **excessive abnormal bone remodeling**, leading to **weak, deformed, and enlarged bones**.

Aetiology:

- Exact cause **unknown**; may have **family history**.
- Possible causes:
  - **Inflammatory** (Paget's suggestion).
  - **Viral infection**.
  - **Autoimmune or vascular disorders**.
  - **Breakdown of normal bone remodeling** (Jaffe).
  - **Paramyxovirus infection** of osteoclasts influenced by **genetic or environmental factors**.

Clinical Features:

- Common in people **over 50 years**. Rare in those <20 years.
- **Male-to-female ratio: 2:1**.
- Common sites: **Maxilla**, skull, femur, tibia, sacrum, pelvis, lumbar spine, and mandible.
- **Bone pain:** Dull, constant, and deep.
- Increased **vascularity:** Bones feel **warm**.
- Enlarged skull may require **larger hat sizes**.
- **Symptoms:**
  - **Fractures, headaches, hearing loss, vision problems**.
  - **Platybasia:** Skull base softening causes neurological issues (e.g., gait difficulties, bowel/bladder issues).
  - **Bowing of femur and tibia**, spinal curvature.
  - **Lion-like facies (Leontiasis ossea)** due to deformity.

Oral Manifestations:

- Jaws commonly involved, especially **maxilla**.
- **Progressive jaw enlargement**; widened alveolar ridges, flattened palate.
- **Loose teeth, spacing issues** due to bone changes.
- Difficulty in **extraction** (cementosis) and risk of **dry socket**.
- **Hypercementosis, ankylosed teeth, submerged teeth**.
- Edentulous patients face **difficulty wearing dentures**.
- **Triangular skull vault**, ill-defined bone ache, atypical facial neuralgias.

Laboratory Findings:

- **Elevated serum alkaline phosphatase** in osteoblastic phase.
- **Normal serum calcium and phosphorus**.
- **Increased urinary hydroxyproline**, NTX, and CTX levels (bone resorption markers).

Radiological Features:

- Phases:
  - **Deossification and softening** (early phase).
  - **Dysplastic reossification** (later phase).
- **Skull X-ray:** Cotton wool appearance.
- Jaws show **poorly defined osteoporosis**, irregular osteoblastic activity.
- Teeth changes: **Hypercementosis, root resorption, displacement, loss of lamina dura**.

Treatment: oral candidiasis

1. **Polyene Agents:**
    - **Nystatin**
    - **Amphotericin B**
2. **Imidazole Agents:**
    - **Clotrimazole**
    - **Ketoconazole**
3. **Triazoles:**
    - **Fluconazole**
    - **Itraconazole**
4. **Other Antifungal Agents:**
    - **Iodoquinol**
5. **Systemic Antifungal Treatment:**
    - For **immunosuppressed** patients, **amphotericin B** and **fluconazole** are used.
6. **Iodoquinol:**
    - Not strictly an antifungal but has antifungal and antibacterial properties.
    - Effective for **angular cheilitis** when compounded with corticosteroids.
7. **Oral Hygiene:**
    - **Improvement of oral hygiene** is essential for prevention and treatment.

Clinical Features:

- **Pseudomembranous oral candidiasis** characterized by **adherent white plaques** resembling **cottage cheese** or **curdled milk**.
- These **white patches** can be **removed** by scraping with a **tongue blade** or rubbing with a **dry gauze sponge**.
- **Infants**: Contracted from **mother's affected vulva, bottle feeding**, or **overcrowded nurseries**.
- **Adults**: Inflammation, erythema, and **eroded areas**.
  - **Deeper invasion** by the organism leads to **ulcerative lesions** upon removal of patches.
- **Adults**: More common in those on **broad-spectrum antibiotics** or **corticosteroid therapy** for prolonged periods.
  - Also occurs in **AIDS** patients.
- **More common in women**.
- **Symptoms**: **Burning sensation** and **foul taste**.
- **White patches** can be wiped off with **wet gauze**, leaving an **erythematous** or **atrophic** area.

Histological Features:

- Histopathologic pattern may vary with the type of **clinical form**.
- **Invasion of mycelia** in **epithelium**, causing **necrosis** of the epithelium.

- **Definition**:
  - Inflammatory cyst lined by **epithelium**, filled with **fluid**.
  - Also called **apical periodontal cyst**, **root-end cyst**, or **periapical cyst**.

Aetiology

- **Infection** from microorganisms or toxins in pulp canal of **non-vital tooth**.
- **Trauma**.
- Originates from **periapical granuloma**:
  - **Rests of Malassez** proliferate in the granuloma.
  - Inner cells undergo **liquefaction necrosis** due to lack of nourishment.
  - Forms a **cystic cavity** in the center of granuloma.

Histopathological Features

- **Lined by stratified squamous epithelium** (variable thickness, rare keratin).
- Presence of:
  - **Hyaline bodies** and **Rushton bodies**
  - **Cholesterol clefts** and **multinucleated giant cells**
- Wall made of **collagen fibers**, **fibroblasts**, and **small blood vessels**.
- **Lumen**: Contains **fluid** with low protein, sometimes **cholesterol** or **keratin**.

Clinical Findings

- Common in **adults**, usually in **males**.
- More frequent in **maxillary anterior region**.
- **Non-vital tooth**, often asymptomatic.
- **Septic tooth or root** may be present.
- **Buccal swelling**:
  - **Round, hard** initially → becomes thin like an **eggshell**.
  - **Crackling sensation** on pressure.
  - Fully resorbed bone → **fluctuation** on pressure.
- **Bluish color** visible under mucous membrane.
- **Infection** causes redness, pain, and **pus discharge** through sinus opening.

Histologic Features

Leukoplakia shows a variety of histologic changes related to **keratinization**, **cellular layer changes**, **epithelium thickness**, and **connective tissue stroma alterations**.

Keratinization pattern:

- Leukoplakia presents **hyperorthokeratinization** or **hyperparakeratinization**, with or without **epithelial dysplasia**.
- There's an abnormal increase in the thickness of the **orthokeratin layer** in areas that are usually keratinized.
- Hyperkeratinization of normally keratinized epithelium or some **parakeratin deposition** in non-keratinized areas is an important criterion.
- **Epithelial dysplasia** is more often associated with hyperkeratinized lesions.

Changes in the cellular layer:

- **Epithelial dysplasia** is the hallmark of changes in leukoplakia.
- Histopathological features include:
    - **Loss of polarity** of basal cells
    - Increased **nuclear-cytoplasmic ratio**

    - Irregular **epithelial stratification**
    - **Cellular pleomorphism**
    - **Nuclear hyperchromatism**
    - Reduced **cellular cohesion**
    - Enlarged **nucleoli**
    - Increased **mitotic figures**

Thickness of the epithelium:

- The thickness changes as **epithelial atrophy** or **acanthosis** in leukoplakia.

Alteration in connective tissue:

- There may be destruction of **collagen fibers** and a **chronic inflammatory cell infiltrate** in the connective tissue stroma.

C). Oral Submucous Fibrosis (OSMF)

Definition: OSMF is a precancerous condition affecting the oral cavity, marked by inflammatory changes in the mucosa leading to fibrosis and loss of elasticity.

Synonyms:

- Atropia idiopathica

- Idiopathic scleroderma of the mouth
- Idiopathic palatal fibrosis
- Sclerosing stomatitis
- Submucosal fibrosis of the palate

Aetiology:

- **Excessive consumption of red chillies**
- **Excessive areca nut chewing**
- Nutritive deficiency
- **Immunological factors**
- Genetic factors
- **Protracted tobacco use**

Pathogenesis: Aetiological factors cause oral mucosa alterations, increasing hypersensitivity reactions, which may lead to malignant transformation (4.5-7.6%).

Clinical Features:

- Affects individuals typically aged 20-40 years
- **Females are affected more than males**
- Commonly seen in buccal mucosa, retromolar area, uvula, tongue
- Initial symptoms: burning sensation, particularly with spicy foods
- Excessive or decreased salivation, defective taste
- **Early stage:** "wet leathery" feeling on palpation
- **Advanced stage:** Blanched, stiff mucosa, trismus, leathery lips, difficulty in lip eversion (microchelia)
- Opaque, marble-like mucosa
- **Difficulty in swallowing** and referred pain in ear
- Circular fibrous bands around the mouth
- **Vertical fibrous beds** on palpation
- Shrunken uvula, "hockeystick appearance"
- **Fibrotic gingiva** with deep pigmentation

Histopathology:

- **Hyperkeratinized, atrophic epithelium**
- Flattening and shortening of rete pegs
- Variable degrees of epithelial dysplasia (nuclear pleomorphism, severe intercellular edema)
- Dilated, congested blood vessels, possible hemorrhage
- **Homogenization and hyalinization** of collagen fibers
- Decreased fibroblasts, narrowed blood vessels due to perivascular fibrosis

Treatment:

- **Cease all causative habits**
- Intralesional injections: **collagenase, corticosteroids, fibrinolysin**
- **Systemic steroids** in severe cases

D). Oral Lichen Planus

Aetiology:

- **Immune-related disorder.**
- **Exacerbations or remissions** often linked to **emotional trauma** (e.g., stress from personal loss, job issues, or exam pressures).

Clinical Features:

- **Common in middle-aged and elderly**, slight female predilection.
- **Areas affected**: buccal mucosa, vestibule, tongue, lips, floor of mouth, palate, gingiva.
- **Symptoms**: Burning sensation.
- **Lesion characteristics**: White and grey **velvety thread-like papules** in **linear, angular, or retiform** arrangements.
- **Wickham's striae**: Tiny white elevated dots at intersections of white lines.

Patterns:

1. Linear
2. Papular
3. Reticular
4. Annular (circular)
5. Vesicular or bullous
6. Erosive or atrophic
7. Hypertrophic

Histopathology:

- **Hyperorthokeratosis** or **parakeratosis**.
- Thickened granular cell layer.
- **Acanthosis** of spinous layer, **saw-tooth appearance** of rete pegs.
- **Necrosis** or **liquefaction degeneration** of basal cell layer.
- **Band-like subepithelial infiltrate** (T cells, histiocytes).
- **Degenerating basal keratinocytes**: rounded or ovoid eosinophilic bodies (Civatte bodies).
- **Max-Joseph spaces**: Histological clefts due to degeneration of basal keratinocytes.

Scanned with OKEN Scanner

clinical features of the Radicular Cyst.

Various periapical lesions:

- **Acute apical periodontitis**
- **Periapical abscess**
- **Periapical granuloma**
- **Periapical cyst**
- **Dentigerous cyst**
- **Periapical scar**
- **Giant cell granuloma**
- **Osteomyelitis**
- **Periapical cemental dysplasia**
- **Langerhans cell disease**

- Definition: